A-A-50457
January 15, 1982

COMMERCIAL ITEM DESCRIPTION

VALVES, RADIATOR, HEATING,
PACKLESS BONNET, 125 PSIG

The General Services Administration has authorized the use of this commercial item description in preference to Types I and II, Styles A and D, Classes 100 and 125, Pattern 2, and Service 1 of Federal Specification WW-V-160.

This commercial item description covers the requirements for Valves, Raditor, Heating, Packless Bonnet. Metallic diaphragm seal for class 125 psig of the angle globe pattern for steam service with female threaded inlet and male threaded outlet connections with union nut and male threaded tailpiece and hand wheel operator.

Salient characteristics: Valves shall be in accordance with table below:

NSN	Size	Type	Style	Class	Pattern	Service
4520-00-720-4735 4520-00-516-4671 4520-00-720-4736	1/2 NPT 3/4 NPT 1 NPT	"II" Packless bonnet	"D" Metallic diaphragm seal	125 psig (working)	2 angle globe	steam

Valves shall be designed for use in steam heating systems at the working pressure and coincident steam saturation temperature applicable to the class specified herein.

Component characteristics. Body and bonnet shall be cast or forged of brass or bronze alloys conforming to ASTM B 61 or B 62. Body castings shall withstand a test pressure equal to twice the rated working pressure. Valves shall be furnished with a phosphor bronze or stainless steel diaphragm seal. The diaphragm shall be either the spring type with aperture or the solid type without aperture. The diaphragm with aperture shall consist of at least two metallic, spring disks which exert a constant force at the aperture against a collar on the stem to maintain a leakproof seal. The diaphragm without aperture shall be a solid, continuous, metallic disk. Valves equipped with solid diaphragms shall include a two-piece stem and an internal, operating-lever assembly. The upper portion of the stem shall bear against the diaphragm and shall be equipped with a free-turning, pressure-bearing button to minimize wear on the diaphragm. Threaded inlet and outlet connections shall have tapered pipe threads conforming to ASME B2.1. Union nuts and tailpieces

FSC 4520

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:42Z
Check the source to verify that this is the current version before use.

shall be ASTM B 61 or B 62 with threads and facings in accordance with contractor's standard. Handwheel suitably attached to stem shall be of molded phenolic or an equivalent plastic or composition material.

Marking. Valves shall be marked in accordance with MSS SP-25.

Certification. The contractor shall certify that the product offered meets the salient characteristics of this description, and that the product conforms to the producer's own drawings, specifications, standards and quality assurance practices, and is the same product sold in the commercial marketplace. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Recovered materials. The manufacturer shall utilize recycled and reclaimed materials to the maximum extent practicable.

Packaging, packing, and marking. Unless otherwise specified in the contract or order, the packaging and packing shall be in accordance with normal commercial practice and shall assure acceptance by common carrier and provide product protection against loss and damage during multiple shipments, handling, and storage. The shipping container shall be in compliance with the National Motor Freight Classification and Uniform Freight Classification. Marking shall be as specified in the contract or order.

NOTES:

ASTM Standards are available from the American Society for Testing Materials, 1916 Race Street, Philadelphia, PA 19103.

MSS Standards are available from the Manufacturers Standardization Society of the Valve and Fittings Industry, 1918 North Fort Myer Drive, Arlington, VA 22209.

MILITARY INTERESTS:

Custodians

Army - ME
Navy - YD
Air Force - 99

Review Activities

DCSC-SSD

User Activities

Army - CE

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS

PREPARING ACTIVITY:

Navy - YD

Project No. 4520-0271-1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:42Z
Check the source to verify that this is the current version before use.